

MEDIATEK

MT8676 Android WiFi

Version: 1.0
Release date: 2024-08-12

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Wenxuan Hu	Official release

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1 WiFi

1.1 Overview

1.1.1 Introduction

This chapter explains the structure of the MT8676 WiFi module and provides guidance on troubleshooting typical WiFi issues. The WiFi chip paired with MT8676 is MT6637.

1.1.2 Abbreviation

Table 1-1. Abbreviations

Abbreviation	Explanation
STA	Station, this term describes a computing device equipped with a wireless network interface, such as a laptop, a WiFi-enabled mobile phone, or a tablet.
AP	Access Point, devices that have the ability to bridge between wireless or wired connections, like wireless routers.
P2P	Peer-to-Peer Connection
BSS	An area that is under the control of an AP
BSSID	MAC Address for AP
SSID	Service Set Identifier
ESS	Extended Service Set, the unique identification of a large-scale virtual Basic Service Set (BSS) is based on multiple BSSs sharing the same SSID.
802.11 series phy layer	802.11b/g/n, 802.11ac, 802.11ax...
2.4G/5G/6G	WiFi Frequency Band
Discovery stage	Passive Scanning (listening Beacon), Active Scanning (Probe request/Probe Response)
Authentication stage	Authentication request, Authentication response
Associate stage	Association request, Association response

Table 1-2. MTK WiFi module name

Module Name	Description
Glue layer (Linux/Android) (OS Adaptation)	Provides access interfaces for OS IOCTL, spin lock, ISR, Main thread, download RAM code and patch common CFG80211 AIS
HIF/HAL	HW Adapter Layer FW own/Driver own, TX data, TX command, RX interrupt
AIS (AdHoc/Infrastructure/Search)	Infra. STA Connection CFG80211 AIS
P2P	P2P Connection CFG80211 P2P P2P Find Phase - P2P Scan (All Channel) - P2P Search (Social Channel)

Module Name	Description
	Remain on Channel Management Frame TX - Off channel TX - Non off channel TX Multiple Interface - Interface add/del/change
AAA (AP(Hotspot) Auth/Assoc)	AP's authentication and association
SAA (Station Auth/Assoc)	STA's authentication and association
MQM (Queue Management)	QoS, Multiple queue control
RLM (Radio Link Management)	Bandwidth, preamble, slot time, OBSS, protection mode,
SCN (Scan)	Queuing of scanning request
SEC (Security)	Encryption, key management
TXM	TX path, CMD queue
RXM	RX path, Re-order buffer, RX AMPDU establishment
CNM (Concurrent Network Management)	Handle channel privilege message

1.2 Architecture/Process Overview

Figure 1-1 shows MT6637 WiFi architecture.

Android Wi-Fi Architecture

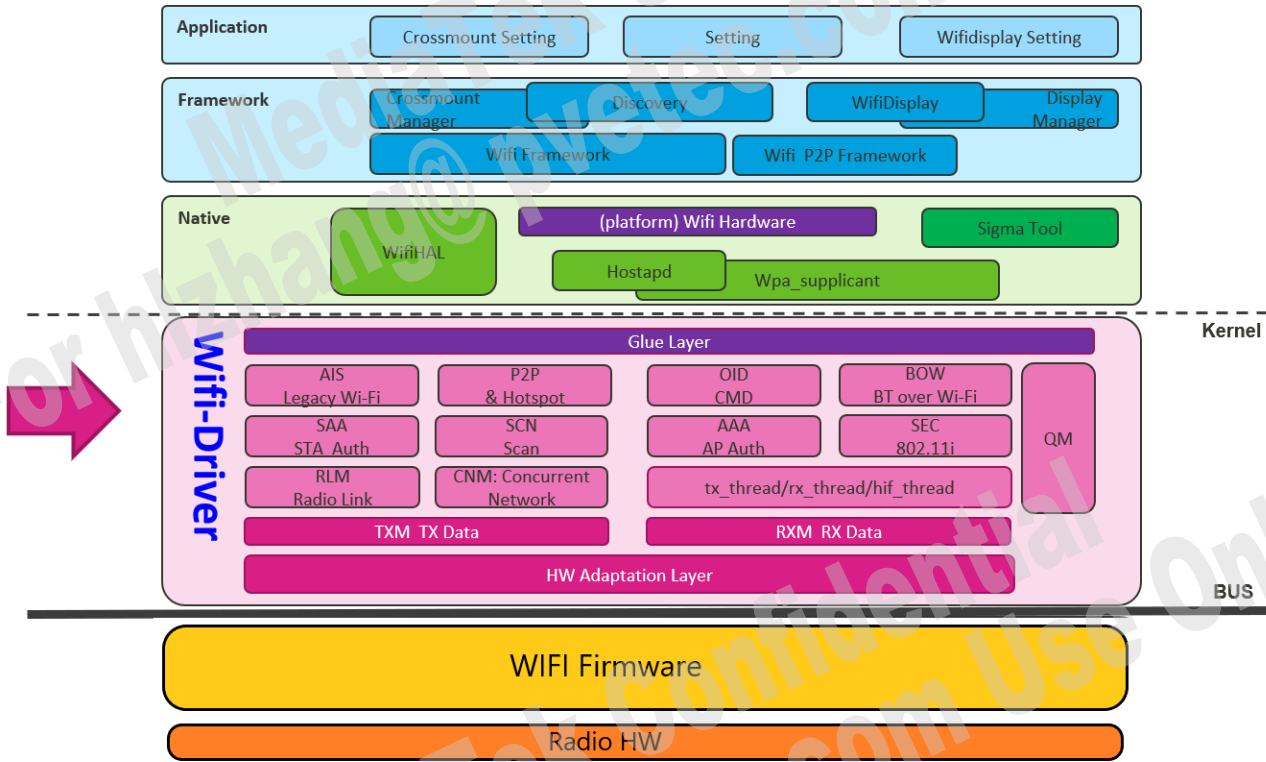


Figure 1-1. WiFi architecture

1.3 Configuration/Customization Guideline

1. NVRAM Customization

WiFi customization typically involves modifying configurations in NVRAM file, which is located in the device as indicated below:

/data/nvram/APCFG/APRDEB/WIFI

NVRAM in code path: /alps/vendor/mediatek/proprietary/custom/\$project/cgen/CFG_WIFI_Default.h

The meaning of the respective field is indicated. Please use meta tool to adjust the relevant parameters.

2. wifi.cfg Customization

Customers can use `wifi.cfg` to configure WiFi features. The public version does not include `wifi.cfg` by default. If customized configuration of WiFi features is required, please create a `wifi.cfg` and add it to the `/vendor/firmware/` of the device.

You can find the default values for configurable parameters in the driver in `wlanInitFeatureOption` and the default value can be modified through `wifi.cfg`.

It is also possible to alter the default parameter values in the firmware by `wifi.cfg`. Specific parameters for modification are provided by the WiFi RD according to the specific functional requirements.

3. Channel list customization

It is typical to customize the channel list with the relevant country code. Customized file path:

/alps/vendor/mediatek/kernel_modules/connectivity/wlan/core/gen4m/mgmt/rlm_domain .c

These are four cases of channel list customization:

case 1: Remove 144 channel of country code US

a. Find `g_u2CountryGroup26` from `COUNTRY_CODE_US`

```
static const uint16_t g_u2CountryGroup26[] = {
    COUNTRY_CODE_AS, COUNTRY_CODE_US
};
```

b. Find supported channel lists of `g_u2CountryGroup1` in `arSupportedRegDomains`

```
{
    (uint16_t *) g_u2CountryGroup26, sizeof(g_u2CountryGroup26) / 2,
    {
        {81, BAND_2G4, CHNL_SPAN_5, 1, 11, FALSE}
        , /* CH_SET_2G4_1_11 */
        {115, BAND_5G, CHNL_SPAN_20, 36, 4, FALSE}
        , /* CH_SET_UNII_LOW_36_48 */
        {118, BAND_5G, CHNL_SPAN_20, 52, 4, TRUE}
        , /* CH_SET_UNII_MID_52_64 */
        {121, BAND_5G, CHNL_SPAN_20, 100, 12, TRUE}
        , /* CH_SET_UNII_WW_100_144 */
        {125, BAND_5G, CHNL_SPAN_20, 149, 8, FALSE}
        , /* CH_SET_UNII_UPPER_149_177 */
    }
}
```

}

c. Modify operating class 121 as shown below

```
{121,BAND_5G,CHNL_SPAN_20,100,11,TRUE}
```

Case 2: Modify will affect supported channel list of all country code in `g_u2CountryGroup1`

a. Define new group

```
static const uint16_t g_u2CountryGroup27[] = {
    COUNTRY_CODE_US
};
```

b. Remove COUNTRY_CODE_US from the original group

c. Define supported channel list of new group

```
{
    (uint16_t *) g_u2CountryGroup27, sizeof(g_u2CountryGroup27) / 2,
    {
        {81, BAND_2G4, CHNL_SPAN_5, 1, 11, FALSE}
        , /* CH_SET_2G4_1_11 */
        {115, BAND_5G, CHNL_SPAN_20, 36, 4, FALSE}
        , /* CH_SET_UNII_LOW_36_48 */
        {118, BAND_5G, CHNL_SPAN_20, 52, 4, TRUE}
        , /* CH_SET_UNII_MID_52_64 */
        {121, BAND_5G, CHNL_SPAN_20, 100, 12, TRUE}
        , /* CH_SET_UNII_WW_100_144 */
        {125, BAND_5G, CHNL_SPAN_20, 149, 8, FALSE}
        , /* CH_SET_UNII_UPPER_149_177 */
    }
}
```

Case3: Remove one supported channel list of US, for example, 52-64

a. Find COUNTRY_CODE_US from country group (`g_u2CountryGroup1`)

```
static const uint16_t g_u2CountryGroup26[] = {
    COUNTRY_CODE_AS, COUNTRY_CODE_US
};
```

b. Find supported channel lists of `g_u2CountryGroup1` in `arSupportedRegDomains`

```
{
    (uint16_t *) g_u2CountryGroup26, sizeof(g_u2CountryGroup26) / 2,
    {
        {81, BAND_2G4, CHNL_SPAN_5, 1, 11, FALSE}
        , /* CH_SET_2G4_1_11 */
        {115, BAND_5G, CHNL_SPAN_20, 36, 4, FALSE}
        , /* CH_SET_UNII_LOW_36_48 */
        {118, BAND_5G, CHNL_SPAN_20, 52, 4, TRUE}
        , /* CH_SET_UNII_MID_52_64 */
        {121, BAND_5G, CHNL_SPAN_20, 100, 12, TRUE}
        , /* CH_SET_UNII_WW_100_144 */
        {125, BAND_5G, CHNL_SPAN_20, 149, 8, FALSE}
        , /* CH_SET_UNII_UPPER_149_177 */
    }
}
```

c. Modify operating class 118 as shown below

```
{118, BAND_NULL, 0, 0, 0, TRUE}
```

Case4: Setting Passive channel

If customer has no special setting, ch52-64 and ch100-140 are passive channel by default.

If customer has special setting, please refer to the following configuration

a. Define `g_u2CountryGroup1_passive` which includes `COUNTRY_CODE_US`

```
static const uint16_t g_u2CountryGroup0_Passive[] = {
    COUNTRY_CODE_US
};
```

b. Add passive channel list to `g_u2CountryGroup1_Passive` in `arSupportedRegDomains_Passive`

```
{
    (uint16_t *) g_u2CountryGroup0_Passive,
    sizeof(g_u2CountryGroup0_Passive) / 2,
    {
        {81, BAND_2G4, CHNL_SPAN_5, 1, 0, FALSE}
        , /* CH_SET_2G4_1_14_NA */
        {82, BAND_2G4, CHNL_SPAN_5, 14, 1, TRUE }
        ,
        {115, BAND_5G, CHNL_SPAN_20, 36, 0, FALSE}
        , /* CH_SET_UNII_LOW_NA */
        {118, BAND_5G, CHNL_SPAN_20, 52, 0, FALSE }
        , /* CH_SET_UNII_MID_52_64 */
        {121, BAND_5G, CHNL_SPAN_20, 100, 0, FALSE }
        , /* CH_SET_UNII_WW_100_140 */
        {125, BAND_5G, CHNL_SPAN_20, 149, 0, FALSE}
        /*
    }
}
```

Parameters introduction:

CHAL_SPAN_5: Channel span

14: Start channel

1: Channel count

The code in red font refers to designating one channel starting from channel 14 as a passive channel, while the other lines indicate setting the number of channels to 0, meaning no passive channels.

1.4 Frequently Asked Questions/Troubleshooting

Customers can perform preliminary analysis on the listed cases in advance. For other questions, please try to catch all the logs at one time according to the requirements and provide them to MediaTek for analysis.

1.4.1 WiFi Certification

WiFi Certification is an initiative of the WiFi Alliance for its members to verify that a WiFi product meets the specification and can interoperate with other WiFi devices. Whether to obtain WiFi certification is determined by the customer.

Typically, devices sold by large manufacturers worldwide will be certified. Figure 1-2 shows the WiFi certification steps.

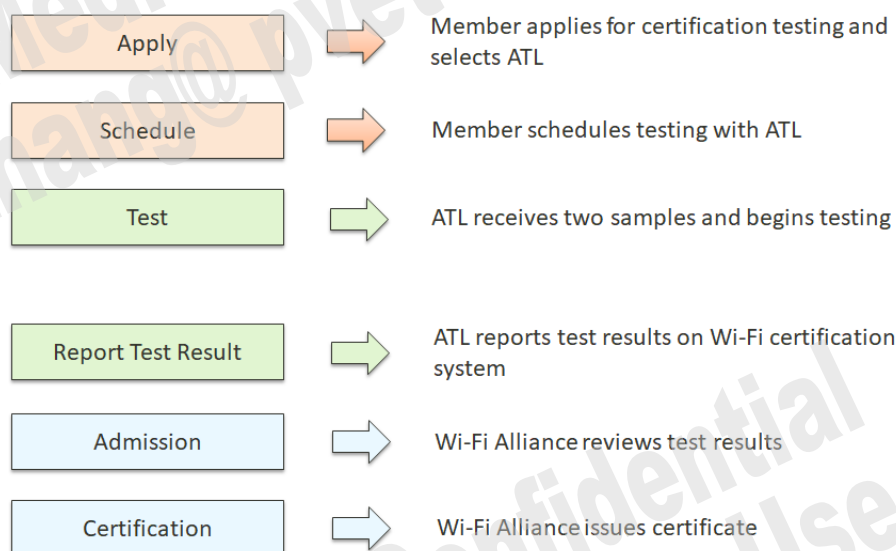


Figure 1-2. WiFi certification steps

The stages where MediaTek assistance is required are as follows:

- **Apply:** When the customer submits the certification application to the alliance, they need to register product information. At this stage, MediaTek may need to assist in confirming the product's capabilities in order to determine which tests need to be conducted.
- **Schedule:** The customer confirms the testing schedule with the laboratory. MediaTek needs to prepare the testing tools before the formal start of the lab, and also needs to ensure that the customer's schedule allows for sufficient time for testing and debugging.
- **Test:** The laboratory conducts testing on the product. During this stage, MediaTek may need to assist in debugging testing issues.

1.4.2 Logs for WiFi Debug

Method to enable WiFi logs (including framework, wpa_supplicant, driver, fw) and how to confirm if they have been successfully enabled.

1. wpa_supplicant log:

```
adb shell "wpa_cli -i wlan0 -g@android:wpa_wlan0 IFNAME=wlan0 LOG_LEVEL DEBUG"
```

2. Framework log:

Settings→About→Build click 6 times, then you will see "you are now a developer!" on the screen Settings→Develop options→Enable Wi-Fi verbose Logging

3. FW log, WiFi driver log and tcpdump:

MediaTek demo board provides DebugloggerUI, with FW logs located in consyslog, WiFi driver logs in mobilelog, and tcpdump in netlog. After collecting the logs, you should pull them out from the /data/debuglogger directory.

4. Change FW log level (UI:DebugloggerUI-->Log Level-->WiFi Firmware Log Level), using the following commands:

Extreme:

```
adb shell "iwpriv wlan0 driver `set_chip EvtDrvLogCatLvl 0xFFFFFFFF`"
```

More:

```
adb shell "iwpriv wlan0 driver `set_chip EvtDrvLogCatLvl 0xFFFFFFFF0F`"
```

Default:

```
adb shell "iwpriv wlan0 driver `set_chip EvtDrvLogCatLvl 0xFFFFFFFF03`"
```

5. Change WiFi driver log level (UI:DebugloggerUI-->Log Level-->WiFi Driver Log Level), using the following commands:

Extreme:

```
adb shell "echo "0xff:0x7f" > /proc/net/wlan/dbgLevel"
```

More:

```
adb shell "echo "0xff:0x3f" > /proc/net/wlan/dbgLevel"
```

Default:

```
adb shell "echo "0xff:0x2f" > /proc/net/wlan/dbgLevel"
```

6. Sniffer log:

Sniffer log is a very important means to analyze WiFi problems by capturing air log. Problems such as connection disconnection, throughput, delay, etc. need to be analyzed with the sniffer log.

After Project kicks off, customer needs to prepare a wireless NIC with the driver and tool installed on the PC for capturing the sniffer log. When providing logs, please provide sniffer logs together to avoid back-and-forth communication about log problems.

1.4.3 Scan

1.4.3.1 Scan Key Log

Scan state machine start:

```
scnFsmSteps: (SCN STATE) [SCAN] TRANSITION: [IDLE] -> [SCANNING]
```

Scan state machine stop:

```
scnFsmSteps: (SCN STATE) [SCAN] TRANSITION: [SCANNING] -> [IDLE]
```

Calculate the time difference between stop and start to see if it is normal.

Scan done event in Kernel log:

```
aisFsmRunEventScanDone: (AIS INFO) ScanDone 1, status(0) native req(1)
```

Scan done reported to supplicant in main log:

```
wpa_supplicant: wlan0: Event SCAN_RESULTS (3) received
```

1.4.3.2 Scan NO AP

Check whether the scan command is delivered by the framework in main log

1. Log:

```
WifiHW: enter -->wifi_send_command cmd=IFNAME=wlan0 SCAN TYPE=ONLY
```

If not, check the framework first. Otherwise, go to next step.

2. Check whether scan state machine has started successfully in the Kernel log.

```
scnFsmSteps: (SCN STATE) [SCAN]TRANSITION: [IDLE] -> [SCANNING]
```

3. Check whether the scan command is sent to FW in Kernel log

```
kalEnqueueCommand: (INIT TRACE) EN-Q CMD TYPE[x] ID[0x03]
```

Whether received scan done event from FW, log:

```
aisFsmRunEventScanDone
```

4. Check whether there are any scanned access points (APs) present in the Kernel log

```
SCANLOG: (SCN INFO) [SCN:600:D2K] Total:12/24
```

Note:

If you need MediaTek's assistance, please provide consyslog, mobileLog, netlog, and sniffer log. The personal computer used to capture the sniffer log must first synchronize its time with the device, and both the IP addresses and MAC addresses of the two sides must be provided. If there is a coexistence with BT 2.4G, please also provide the picus log (clearly indicating the time of the issue).

1.4.4 STA Connect

1. Check connect policy, with the filter condition: mtk_cfg80211_connect

```
mtk_cfg80211_connect: (REQ INFO) [wlan] mtk_cfg80211_connect 00000000e0a61a62 61
auth_type=0 flags=0x40 wlanoidSetConnect: (INIT INFO) ucBssIndex 0, ssid test_ap, bssid
00:00:00:00:00:00, bssid_hint ce:84:f6:15:4f:4d, conn policy 4, disc reason 4, freqInMHZ
5805
```

2. Check whether AIS State flow works normally, with the filter condition: (AIS STATE)

```
aisFsmSteps: (AIS STATE) [AIS0][0] TRANSITION: [IDLE] -> [IDLE]
aisFsmSteps: (AIS INFO) eReqType=1
aisFsmSteps: (AIS STATE) [AIS0][0] TRANSITION: [IDLE] -> [SEARCH]
aisFsmSteps: (AIS STATE) [AIS0][0] TRANSITION: [SEARCH] -> [REQ_CHANNEL_JOIN]
aisFsmSteps: (AIS STATE) [AIS0][0] TRANSITION: [REQ_CHANNEL_JOIN] -> [JOIN]
aisFsmSteps: (AIS STATE) [AIS0][0] TRANSITION: [JOIN] -> [NORMAL_TR]
```

3. Check whether SAA State flow works normally, with the filter condition: (SAA STATE)

```
saaFsmSteps: (SAA STATE) [SAA]TRANSITION: [AA_IDLE] -> [SAA_SEND_AUTH1]
saaFsmSteps: (SAA STATE) [SAA]TRANSITION: [SAA_SEND_AUTH1] -> [SAA_WAIT_AUTH2]
saaFsmSteps: (SAA STATE) [SAA]TRANSITION: [SAA_WAIT_AUTH2] -> [SAA_SEND_ASSOC1]
```

```
saaFsmSteps: (SAA STATE) [SAA] TRANSITION: [SAA_SEND_ASSOC1] -> [SAA_WAIT_ASSOC2]
saaFsmSteps: (SAA STATE) [SAA] TRANSITION: [SAA_WAIT_ASSOC2] -> [AA_IDLE]
```

4. Check whether the selected AP is connected, with the filter condition: netif_carrier_on
5. Check whether the AIS State is set to normal TR, with the filter condition: (AIS STATE)

```
aisFsmSteps: (AIS STATE) [AIS0] TRANSITION: [JOIN] -> [NORMAL_TR]
```

6. Confirm the EAP behavior, with the filter condition: EAPOL

```
statsParsePktInfo: (RX INFO) <RX> EAPOL: key, M1, KeyInfo 0x008a, SSN:0
statsParsePktInfo: (TX INFO) <TX> EAPOL: key, M2, KeyInfo 0x010a SeqNo:2
statsParsePktInfo: (RX INFO) <RX> EAPOL: key, M3, KeyInfo 0x13ca, SSN:1
statsParsePktInfo: (TX INFO) <TX> EAPOL: key, M4, KeyInfo 0x030a SeqNo:3
```

7. Confirm the DHCP behavior, with the filter condition: DHCP

The Kernel log can determine whether the DHCP is successfully sent and received

Ensure that the DHCP IPID and MsgType <TX/RX> are paired

TX status are all 0, indicating success.

Note:

If you need MediaTek's assistance, please provide consyslog, mobileLog, netlog, and sniffer log. The personal computer used to capture the sniffer log must first synchronize its time with the device, and both the IP addresses and MAC addresses of the two sides must be provided. If there is a coexistence with BT 2.4G, please also provide the picus log (clearly indicating the time of the issue).

1.4.5 Throughput

- Network throughput is influenced by various factors, including human operations, environment, protocol limitations, hardware, software, etc. The more conditions collected, the shorter the time to resolve the issue.
- To avoid the impact of environmental factors and misalignment of testing tools, it is recommended to conduct throughput testing in a shielded room. It is suggested to use the iperf tool in a Linux environment for testing.
- Test devices of the same level in an equivalent environment, observe the differences, and replace access points (APs) to verify the phenomenon.
- Ensure to conduct CTIA mode testing to avoid interference from online scanning, power-saving modes, etc.
- Throughput testing should ideally be pure WiFi testing, so be sure to turn off Bluetooth (BT).
- If there is a coexistence with Bluetooth, the scenario must be clearly defined, and picus logs should be provided

Note: If you need MediaTek's assistance, please provide consyslog, mobileLog, netlog, and sniffer logs (Synchronize the time between the DUT and the pc that to capture sniffer log, and both IP and mac addresses must be provided). If it coexists with BT 2.4G, picus log is provided synchronously.

1.4.6 MT6637 Chip Reset

If the WiFi driver or WiFi FW is abnormal and cannot be restored, the chip reset is triggered. Before the chip resets, the system uses the coredump mechanism to collect some status on the scene.

In this case, the coredump file and consyslog are important analysis methods, and must be provided.

Coredump mechanism

- 1. Through netlink, the driver sends data to the native
- 2. Native get EMI by mmap.

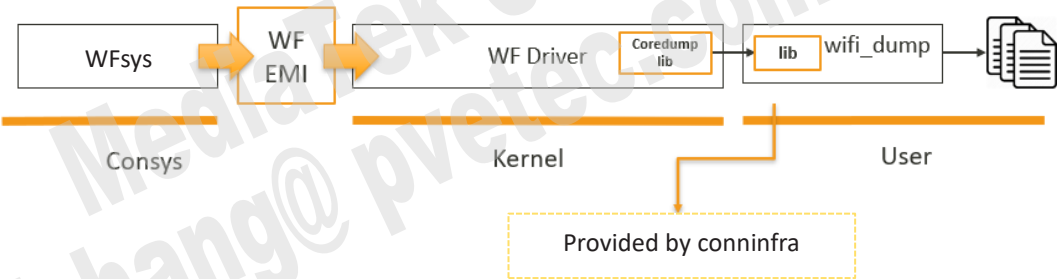


Table 1-3. Coredump file path

	BT	WiFi
Assert_type	combo_bt	combo_wifi
Dump 存放目录 (可在 native 端客制化)	/data/vendor/connsyslog/bt	/data/vendor/connsyslog/wifi
Dump file list	combo_t32.cmm SYS_WCN_ISSUE_INFO SYS_WCN_EMI_DUMP SYS_WCN_MCIF_EMI_DUM SYS_WCN__ILM_DUMP SYS_WCN__DLM_DUMP SYS_WCN_SRAM_DUMP	combo_t32.cmm SYS_WCN_ISSUE_INFO SYS_WCN_EMI_DUMP SYS_WCN_MCIF_EMI_DUM SYS_WCN__ROM_DUMP SYS_WCN__ILM_DUMP SYS_WCN_MDLM_DUMP SYS_WCN_WDLM_DUMP SYS_WCN_SRAM_DUMP

Follow the steps below to check whether the coredump mechanism is enabled.

- 1. Check whether coredump mode is set as 1 or 2.

```
getprop persist.vendor.connsys.coredump.mode
```

- 2. If not, set it as 1 or 2, then reboot.

```
setprop persist.vendor.connsys.coredump.mode 2
```

- 3. After the restart, check whether the coredump mode is set to the specified value and whether WiFi dump runs.

```
auto8676p1_64_bsp:/ # getprop | grep coredump
getprop | grep coredump
[persist.vendor.connsys.coredump.mode]: [2]

auto8676p1_64_bsp:/ # ps -A |grep wifi_dump
ps -A |grep wifi_dump
system          1121      1   10788624   5268 poll_schedule_timeout 0 S wifi_dump
```

- 4. If the WiFi dump runs, enable the WiFi first and manually trigger the chip reset to check whether the coredump file can be generated.

```
auto8676p1_64_bsp:/ # echo 0xDB9DB9 > /proc/driver/conninfra_dbg
auto8676p1_64_bsp:/ # echo 0x1 > /proc/driver/conninfra_dbg
```

Check whether a coredump file is generated.

```
auto8676p1_64_bsp:/ # ls data/vendor/connsyslog/wifi
ls data/vendor/connsyslog/wifi
combo_t32_20240602073036.CI_M   combo_t32_20240602073036._ROM
combo_t32_20240602073036.MDLM  combo_t32_20240602073036.cmm
combo_t32_20240602073036.SRAM  combo_t32_20240602073036.emi
combo_t32_20240602073036.WDLM  combo_t32_20240602073036_issue_info.xml
combo_t32_20240602073036._ILM  combo_t32_20240602073036_mcif.emi
```

5. If the coredump file can be generated successfully, you can begin to capture the problem log.

After the fault is detected, provide the coredump file in addition to the debuglogger and the db file generated by the AEE.

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